

// 10 Dhadge Ishwari

Practical 1

// A Java program for a Server

```
import java.net.*;
import java.io.*;
public class Server
{
    //initialize socket and input stream
    private Socket socket = null;
    private ServerSocket server = null;
    private DataInputStream in = null;
    // constructor with port
    public Server(int port)
    {
        // starts server and waits for a connection
        try
        {
            server = new ServerSocket(port);
            System.out.println("Server started");
            System.out.println("Waiting for a client ...");
            socket = server.accept();
            System.out.println("Client accepted");
            // takes input from the client socket
            in = new DataInputStream(
                new BufferedInputStream(socket.getInputStream()));
            String line = "";
            // reads message from client until "Over" is sent
            while (!line.equals("Over"))
            {
                try
                {
                    {
```

```

line = in.readUTF();
System.out.println(line);
}
catch(IOException i)
{
System.out.println(i);
}
}
System.out.println("Closing connection");
// close connection
socket.close();
in.close();
}
catch(IOException i)
{
System.out.println(i);
}
}
public static void main(String args[])
{
Server server = new Server(5000);
}

```

// A Java program for a Client

```

import java.io.*;
import java.net.*;

public class Client {
// initialize socket and input output streams
private Socket socket = null;
private BufferedReader d = null;
private InputStream input = null;
private DataOutputStream out = null;

```

```

// constructor to put ip address and port
public Client(String address, int port)
{
    // establish a connection
    try {
        socket = new Socket(address, port);
        System.out.println("Connected");
        System.out.println("Done with 1st program of DS");
        // takes input from terminal
        d = new BufferedReader(new InputStreamReader(System.in));
        // sends output to the socket
        out = new DataOutputStream(
            socket.getOutputStream());
    }
    catch (UnknownHostException u) {
        System.out.println(u);
        return;
    }
    catch (IOException i) {
        System.out.println(i);
        return;
    }
    // string to read message from input
    String line = "";
    // keep reading until "Over" is input
    while (!line.equals("Over")) {
        try {
            line = d.readLine();
            out.writeUTF(line);
        }
        catch (IOException i) {

```

```

System.out.println(i);
}
}

// close the connection

try {
input.close();

out.close();

socket.close();
}

catch (IOException i) {

System.out.println(i);
}
}

public static void main(String args[])
{

Client client = new Client("127.0.0.1", 5000);

}

}

```

```

3.2.3.jar: command not found
itgtt-vostro-3710:~/Desktop$ cd -
itgtt-vostro-3710:~/Documents
itgtt-vostro-3710:~/Documents$ cd Ishwari
itgtt-vostro-3710:~/Documents/Ishwari$ javac Search.java
itgtt-vostro-3710:~/Documents/Ishwari$ javac SearchQuery.java
itgtt-vostro-3710:~/Documents/Ishwari$ rmic SearchQuery
Warning: generation and use of skeletons and static stubs for JRMP
is deprecated. Skeletons are unnecessary, and static stubs have
been superseded by dynamically generated stubs. Users are
encouraged to migrate away from using rmic to generate skeletons and static
stubs. See the documentation for java.rmi.server.UnicastRemoteObject.
itgtt-vostro-3710:~/Documents/Ishwari$ rmic registry &
[1] 7681
itgtt-vostro-3710:~/Documents/Ishwari$ javac Server.java
itgtt-vostro-3710:~/Documents/Ishwari$ java Server
Server started
waiting for a client ...
Client accepted

3.2.3.jar: command not found
itgtt-vostro-3710:~/Documents$ cd Ishwari
itgtt-vostro-3710:~/Documents/Ishwari$ javac Client.java
itgtt-vostro-3710:~/Documents/Ishwari$ java Client
Connected
Done with 1st program of DS

```

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Practical 2

code as Calc.idl]

module CalcApp

{

interface Calc

{

exception DivisionByZero {};

float sum(in float a, in float b);

float div(in float a, in float b) raises (DivisionByZero);

float mul(in float a, in float b);

float sub(in float a, in float b);

};

};

//CalcServer.java

import CalcApp.*;

import CalcApp.CalcPackage.DivisionByZero;

import org.omg.CosNaming.*;

import org.omg.CosNaming.NamingContextPackage.*;

import org.omg.CORBA.*;

import org.omg.PortableServer.*;

import java.util.Properties;

class CalcImpl extends CalcPOA {

@Override

public float sum(float a, float b) {

return a + b;

}

@Override

public float div(float a, float b) throws DivisionByZero {

if (b == 0) {

throw new CalcApp.CalcPackage.DivisionByZero();

```

    } else {
        return a / b;
    }
}

@Override
public float mul(float a, float b) {
    return a * b;
}

@Override
public float sub(float a, float b) {
    return a - b;
}

private ORB orb;

public void setORB(ORB orb_val) {
    orb = orb_val;
}

}

public class CalcServer {
    public static void main(String args[]) {
        try {
            // create and initialize the ORB
            ORB orb = ORB.init(args, null);

            // get reference to rootpoa & activate the POAManager
            POA rootpoa = POAHelper.narrow(orb.resolve_initial_references("RootPOA"));
            rootpoa.the_POAManager().activate();

            // create servant and register it with the ORB
            CalcImpl helloImpl = new CalcImpl();
            helloImpl.setORB(orb);

            // get object reference from the servant
            org.omg.CORBA.Object ref = rootpoa.servant_to_reference(helloImpl);
            Calc href = CalcHelper.narrow(ref);

```

```

// get the root naming context
// NameService invokes the name service
org.omg.CORBA.Object objRef = orb.resolve_initial_references("NameService");
// Use NamingContextExt which is part of the Interoperable
// Naming Service (INS) specification.
NamingContextExt ncRef = NamingContextExtHelper.narrow(objRef);
// bind the Object Reference in Naming
String name = "Calc";
NameComponent path[] = ncRef.to_name(name);
ncRef.rebind(path, href);
System.out.println("Ready..");
// wait for invocations from clients
orb.run();
} catch (Exception e) {
System.err.println("ERROR: " + e);
e.printStackTrace(System.out);
}
System.out.println("Exiting ...");
}}

```

//CalcClient.java

```

import java.io.BufferedReader;
import java.io.IOException;
import java.io.InputStreamReader;
import CalcApp.*;
import CalcApp.CalcPackage.DivisionByZero;
import org.omg.CosNaming.*;
import org.omg.CosNaming.NamingContextPackage.*;
import org.omg.CORBA.*;
import static java.lang.System.out;
public class CalcClient {

```

```

static Calc calcImpl;

static BufferedReader br = new BufferedReader(new InputStreamReader(System.in));

public static void main(String args[]) {
    try {
        // create and initialize the ORB
        ORB orb = ORB.init(args, null);

        // get the root naming context
        org.omg.CORBA.Object objRef = orb.resolve_initial_references("NameService");

        // Use NamingContextExt instead of NamingContext. This is
        // part of the Interoperable naming Service.
        NamingContextExt ncRef = NamingContextExtHelper.narrow(objRef);

        // resolve the Object Reference in Naming
        String name = "Calc";

        calcImpl = CalcHelper.narrow(ncRef.resolve_str(name));

        // System.out.println(calcImpl);

        while (true) {
            out.println("1. Sum");
            out.println("2. Sub");
            out.println("3. Mul");
            out.println("4. Div");
            out.println("5. exit");
            out.println("--");
            out.println("choice: ");

            try {
                String opt = br.readLine();

                if (opt.equals("5")) {
                    break;
                } else if (opt.equals("1")) {
                    out.println("a+b= " + calcImpl.sum(getFloat("a"), getFloat("b")));
                } else if (opt.equals("2")) {
                    out.println("a-b= " + calcImpl.sub(getFloat("a"), getFloat("b")));
                }
            }
        }
    }
}

```

```

    } else if (opt.equals("3")) {
        out.println("a*b= " + calcImpl.mul(getFloat("a"), getFloat("b")));
    } else if (opt.equals("4")) {
        try {
            out.println("a/b= " + calcImpl.div(getFloat("a"), getFloat("b")));
        } catch (DivisionByZero de) {
            out.println("Division by zero!!!");
        }
    }
    } catch (Exception e) {
        out.println("===");
        out.println("Error with numbers");
        out.println("===");
    }
    out.println("");
}

//calcImpl.shutdown();
} catch (Exception e) {
    System.out.println("ERROR : " + e);
    e.printStackTrace(System.out);
}
}

static float getFloat(String number) throws Exception {
    out.print(number + ": ");
    return Float.parseFloat(br.readLine()); } }

```

[illegible]

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//Practical 3

```
#include <stdio.h>
#include "mpi.h"
int main(int argc, char* argv[])
{
    int rank, size, len;
    MPI_Init(&argc, &argv);
    MPI_Comm_rank(MPI_COMM_WORLD, &rank);
    MPI_Comm_size(MPI_COMM_WORLD, &size);
    printf("Hello, world, I am %d of %d\n",rank, size);
    MPI_Finalize();
    return 0;
}
```

Compile the program

```
nllabc2d22@nllabc2d-22:~/opt/openmpi/bin$ mpicc hello.c
```

Execute the program using 2 cores

```
nllabc2d22@nllabc2d-22:~/opt/openmpi/bin$ mpirun -np 2 ./a.out
```

Hello, world, I am 0 of 2

Hello, world, I am 1 of 2

Execute the program using 4 cores

```
nllabc2d22@nllabc2d-22:~/opt/openmpi/bin$ mpirun -np 4 ./a.out
```

Hello, world, I am 0 of 4

Hello, world, I am 3 of 4

Hello, world, I am 1 of 4

Hello, world, I am 2 of 4

Program to transfer data from core 0 to core 1.

```
#include <stdio.h>
#include "mpi.h"
int main(int argc, char* argv[])
{
```

```

int rank, size, len;

int num=10;

MPI_Init(&argc, &argv);

MPI_Comm_rank(MPI_COMM_WORLD, &rank);

MPI_Comm_size(MPI_COMM_WORLD, &size);

if(rank == 0)

{
printf("Sending message containing: %d from rank %d\n", num,rank);

MPI_Send(&num, 1, MPI_INT, 1, 1, MPI_COMM_WORLD);

}

else

{
printf(" at rank %d\n",rank);

MPI_Recv(&num, 1, MPI_INT, 0, 1, MPI_COMM_WORLD, MPI_STATUS_IGNORE);

printf("Received message containing: %d at rank %d\n", num,rank);

}

MPI_Finalize();

return 0;

}

Sending message containing: 10 from rank 0

at rank 1

at rank 3

Received message containing: 10 at rank 1

at rank 2

/***** The cores 2 and will be in waiting mode ... Press Ctrl+z to end the execution

*****/Assignment program: Add 20 numbers in an array using 4 cores

#include <stdio.h>

#include "mpi.h"

int main(int argc, char* argv[])

{

int rank, size;

```

```

int num[20]; //N=20, n=4

MPI_Init(&argc, &argv);

MPI_Comm_rank(MPI_COMM_WORLD, &rank);

MPI_Comm_size(MPI_COMM_WORLD, &size);

for(int i=0;i<20;i++)
num[i]=i+1;

if(rank == 0){

int s[4];

printf("Distribution at rank %d \n", rank);

for(int i=1;i<4;i++)

MPI_Send(&num[i*5], 5, MPI_INT, i, 1, MPI_COMM_WORLD); //N/n i.e. 20/4=5

int sum=0, local_sum=0;

for(int i=0;i<5;i++)

{

local_sum=local_sum+num[i];

}

for(int i=1;i<4;i++)

{

MPI_Recv(&s[i], 1, MPI_INT, i, 1, MPI_COMM_WORLD, MPI_STATUS_IGNORE);

}

printf("local sum at rank %d is %d\n", rank,local_sum); sum=local_sum; for(int i=1;i<4;i++) printf("local sum at rank %d is %d\n", rank,local_sum);

sum=local_sum;

for(int i=1;i<4;i++)

sum=sum+s[i];

printf("final sum = %d\n\n",sum);

} else {

int k[5];

MPI_Recv(k, 5, MPI_INT, 0, 1, MPI_COMM_WORLD, MPI_STATUS_IGNORE);

int local_sum=0;

for(int i=0;i<5;i++)

```

```

{
local_sum=local_sum+k[i];
}

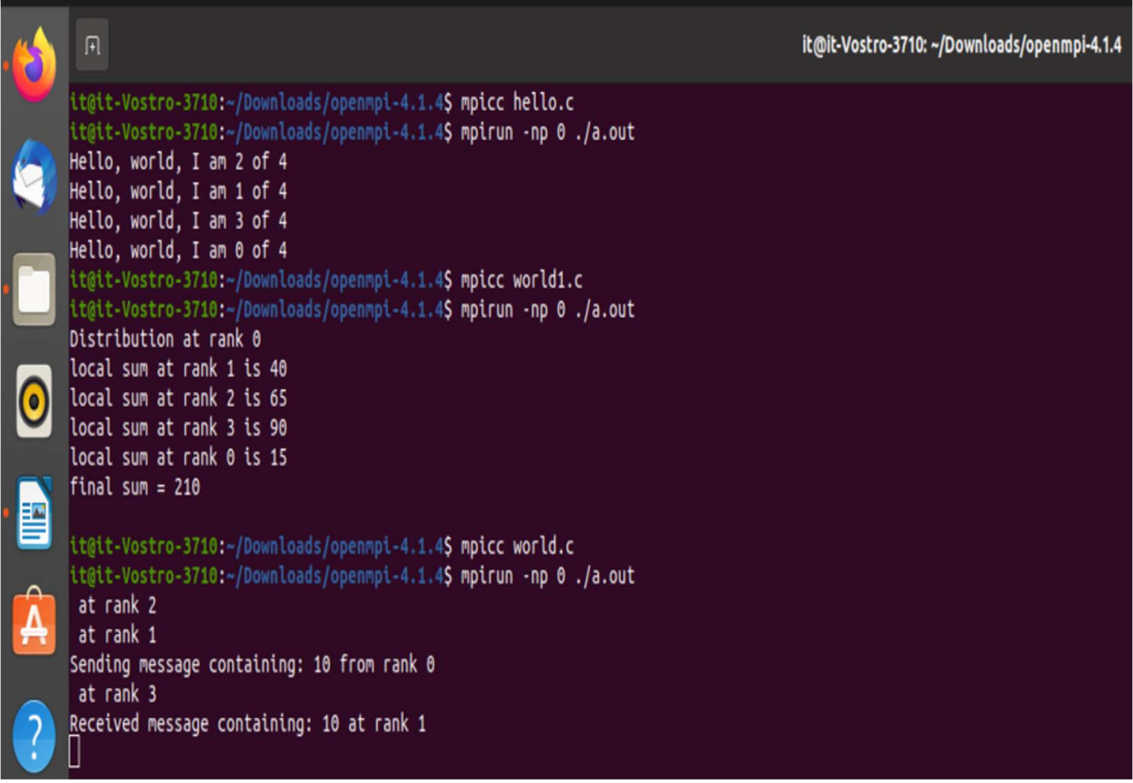
printf("local sum at rank %d is %d\n", rank, local_sum);

MPI_Send(&local_sum, 1, MPI_INT, 0, 1, MPI_COMM_WORLD);
}

MPI_Finalize();

return 0;
}

```



A terminal window titled "it@it-Vostro-3710: ~/Downloads/openmpi-4.1.4" showing the execution of an MPI program. The window has a dark background with a sidebar on the left containing icons for various applications. The terminal output shows the compilation and execution of two programs, "hello.c" and "world.c".

```

it@it-Vostro-3710:~/Downloads/openmpi-4.1.4$ mpicc hello.c
it@it-Vostro-3710:~/Downloads/openmpi-4.1.4$ mpirun -np 4 ./a.out
Hello, world, I am 2 of 4
Hello, world, I am 1 of 4
Hello, world, I am 3 of 4
Hello, world, I am 0 of 4
it@it-Vostro-3710:~/Downloads/openmpi-4.1.4$ mpicc world1.c
it@it-Vostro-3710:~/Downloads/openmpi-4.1.4$ mpirun -np 4 ./a.out
Distribution at rank 0
local sum at rank 1 is 40
local sum at rank 2 is 65
local sum at rank 3 is 90
local sum at rank 0 is 15
final sum = 210
it@it-Vostro-3710:~/Downloads/openmpi-4.1.4$ mpicc world.c
it@it-Vostro-3710:~/Downloads/openmpi-4.1.4$ mpirun -np 4 ./a.out
at rank 2
at rank 1
Sending message containing: 10 from rank 0
at rank 3
Received message containing: 10 at rank 1

```

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//Practical 4

```
import java.util.ArrayList;

public class BerkeleyClockSync {

    public static void main(String[] args) {

        // Initialize the system clocks

        int[] systemClocks = { 10, 12, 13, 11, 14 };

        int masterClock = 0;

        // Print the initial system clocks

        System.out.print("System clocks: ");

        for (int clock : systemClocks) {

            System.out.print(clock + " ");

        }

        System.out.println();

        // Calculate the average system clock

        int sum = 0;

        for (int clock : systemClocks) {

            sum += clock;

        }

        int averageClock = sum / systemClocks.length;

        // Calculate the time difference for each system clock

        ArrayList<Integer> timeDifferences = new ArrayList<>();

        for (int clock : systemClocks) {

            timeDifferences.add(averageClock - clock);

        }

        // Calculate the time adjustment for the master clock

        int timeAdjustment = 0;

        for (int difference : timeDifferences) {

            timeAdjustment += difference;

        }

        timeAdjustment /= timeDifferences.size();

    }

}
```

```
// Update the master clock

masterClock = averageClock - timeAdjustment;

// Print the updated system clocks and master clock

System.out.print("Updated system clocks: ");

for (int clock : systemClocks) {

    System.out.print((clock - timeAdjustment) + " ");

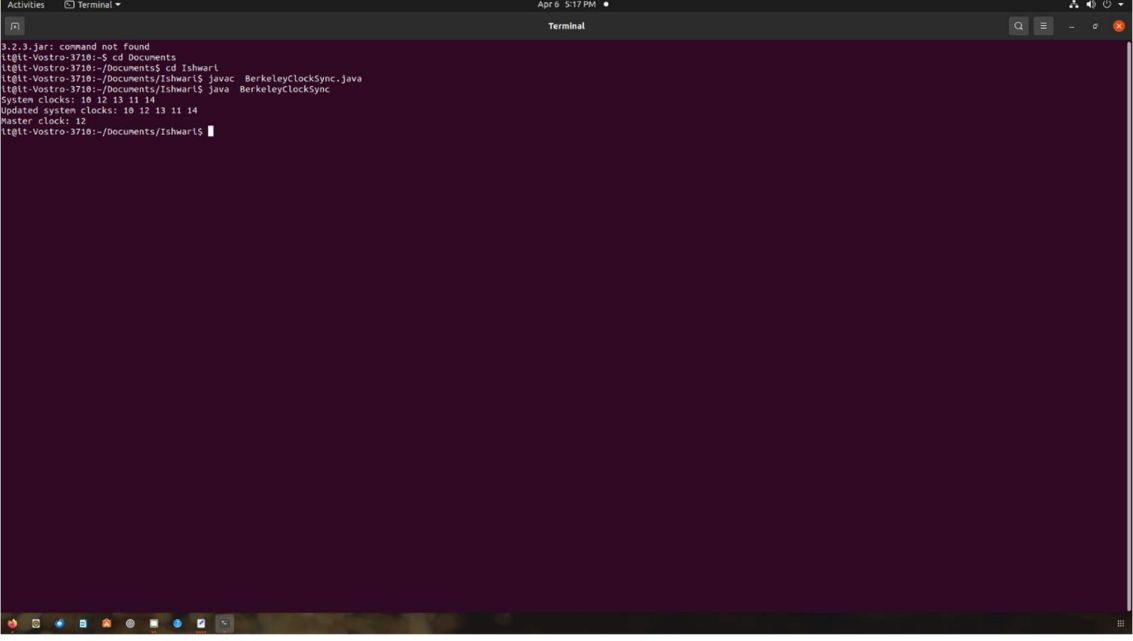
}

System.out.println();

System.out.println("Master clock: " + masterClock);

}

}
```

A screenshot of a Linux terminal window. The window title is "Terminal" and the date/time is "Apr 6 5:17 PM". The terminal shows the following sequence of commands and output:
1. Command: `3.2.3.jar: command not found`
2. Command: `it@lt-vostro-3710:~$ cd Documents`
3. Command: `it@lt-vostro-3710:~/Documents$ cd Ishwar1`
4. Command: `it@lt-vostro-3710:~/Documents/Ishwar1$ javac BerkeleyClockSync.java`
5. Command: `it@lt-vostro-3710:~/Documents/Ishwar1$ java BerkeleyClockSync`
6. Output: `System clocks: 10 12 13 11 14`
7. Output: `Updated system clocks: 10 12 13 11 14`
8. Output: `Master clock: 12`
The terminal prompt is `it@lt-vostro-3710:~/Documents/Ishwar1$`.

```
3.2.3.jar: command not found
it@lt-vostro-3710:~$ cd Documents
it@lt-vostro-3710:~/Documents$ cd Ishwar1
it@lt-vostro-3710:~/Documents/Ishwar1$ javac BerkeleyClockSync.java
it@lt-vostro-3710:~/Documents/Ishwar1$ java BerkeleyClockSync
System clocks: 10 12 13 11 14
Updated system clocks: 10 12 13 11 14
Master clock: 12
it@lt-vostro-3710:~/Documents/Ishwar1$
```

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//Practical 5

```
import java.io.*;
import java.util.*;

class tokenring {

    public static void main(String args[]) throws Throwable {

        Scanner scan = new Scanner(System.in);

        System.out.println("Enter the num of nodes:");

        int n = scan.nextInt();

        int m = n - 1;

        // Decides the number of nodes forming the ring

        int token = 0;

        int ch = 0, flag = 0;

        for (int i = 0; i < n; i++) {

            System.out.print(" " + i);

        }

        System.out.println(" " + 0);

        do{

            System.out.println("Enter sender:");

            int s = scan.nextInt();

            System.out.println("Enter receiver:");

            int r = scan.nextInt();

            System.out.println("Enter Data:");

            int a;

            a = scan.nextInt();

            System.out.print("Token passing:");

            for (int i = token, j = token; (i % n) != s; i++, j = (j + 1) % n) {

                System.out.print(" " + j + "->");

            }

            System.out.println(" " + s);

            System.out.println("Sender " + s + " sending data: " + a);
```

```

    for (int i = s + 1; i != r; i = (i + 1) % n)

System.out.println("data " + a + " forwarded by " + i);

    }

System.out.println("Receiver " + r + " received data: " + a + "\n");

token = s;

do{

try {

if( flag == 1)

System.out.print("Invalid Input!!...");

System.out.print("Do you want to send again?? enter 1 for Yes and 0 for No : ");

ch = scan.nextInt();

if( ch != 1 && ch != 0 )

flag = 1;

else

flag = 0;

} catch (InputMismatchException e){

System.out.println("Invalid Input");

}

}while( ch != 1 && ch != 0 );

}while( ch == 1 );

}}

```

```

1.2.3.jar: command not found
it@lt-vostro-3710:~$ cd Documents
it@lt-vostro-3710:~/Documents$ cd ishwari
it@lt-vostro-3710:~/Documents/ishwari$ javac tokenring.java
it@lt-vostro-3710:~/Documents/ishwari$ java tokenring
Enter the num of nodes:
3
0 1 2 0
Enter sender:
1
Enter receiver:
2
Enter Data:
4
Token passing: 0-> 1
Sender 1 sending data: 4
Receiver 2 received data: 4
Do you want to send again?? enter 1 for Yes and 0 for No : 1
Enter sender:
2
Enter receiver:
1
Enter Data:
5
Token passing: 1-> 2
Sender 2 sending data: 5
data 5 forwarded by 3
Receiver 1 received data: 5
Do you want to send again?? enter 1 for Yes and 0 for No : 0
it@lt-vostro-3710:~/Documents/ishwari$

```

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//Prcatical 6

```
import java.io.InputStream;
import java.io.PrintStream;
import java.util.Scanner;
public class Bully
{
    static boolean[] state = new boolean[5];
    int coordinator;
    public static void up(int up)//4
    {
        if (state[up - 1])// 0 1 2 3 4
        System.out.println("process" + up + "is already up");
    }
    else
    {
        int i;
        Bully.state[up - 1] = true;
        System.out.println("process " + up + "held election");
        for (i = up; i < 5; ++i)
        {
            System.out.println("election message sent from process" + up + "to process" + (i + 1));
        }
        for (i = up + 1; i <= 5; ++i)
        {
            if (!state[i - 1]) continue;
            System.out.println("alive message send from process" + i + "to process" + up);
            break;
        }
    }
}
```

```

public static void down(int down)
{
    if (!state[down - 1])
    {
        System.out.println("process " + down + "is already down.");
    }
    else
    {
        Bully.state[down - 1] = false;
    }
}

public static void mess(int mess)
{
    if (state[mess - 1])
    {
        if (state[4])
        {
            System.out.println("OK");
        }
        else if (!state[4])
        {
            int i;
            System.out.println("process" + mess + "election");
            for (i = mess; i < 5; ++i)
            {
                System.out.println("election send from process" + mess + "to process " + (i + 1));
            }
            for (i = 5; i >= mess; --i)
            {
                if (!state[i - 1]) continue;
                System.out.println("Coordinator message send from process" + i + "to all");
            }
        }
    }
}

```

```

break;
}
}
}
else
{
    System.out.println("Process" + mess + "is down");
}
}

public static void main(String[] args)
{
    int choice;

    Scanner sc = new Scanner(System.in);
    for (int i = 0; i < 5; ++i)
    {
        Bully.state[i] = true;
    }

    System.out.println("5 active process are:");
    System.out.println("Process up = p1 p2 p3 p4 p5");
    System.out.println("Process 5 is coordinator");
    do
    {
        System.out.println(".....");
        System.out.println("1 up a process.");
        System.out.println("2.down a process");
        System.out.println("3 send a message");
        System.out.println("4.Exit");
        choice = sc.nextInt();
        switch (choice)
        {
            case 1:

```

```

{
    System.out.println("bring proces up");
    int up = sc.nextInt();
    if (up == 5)
    {
        System.out.println("process 5 is co-ordinator");
        Bully.state[4] = true;
        break;
    }
    Bully.up(up);
    break;
}
case 2:
{
    System.out.println("bring down any process.");
    int down = sc.nextInt();
    Bully.down(down);
    break;
}
case 3:
{
    System.out.println("which process will send message");
    int mess = sc.nextInt();
    Bully.mess(mess);
}
}
} while (choice != 4);
}
}

```

2. Code (save this code as Ring.java)

```
import java.util.Scanner;
```

```

public class Ring
{
    public static void main(String[] args)
    {
        // TODO Auto-generated method stub
        int temp, i, j;
        char str[] = new char[10];
        Rr proc[] = new Rr[10];
        // object initialisation
        for (i = 0; i < proc.length; i++)
            proc[i] = new Rr();
        // scanner used for getting input from console
        Scanner in = new Scanner(System.in);
        System.out.println("Enter the number of process : ");
        int num = in.nextInt();
        // getting input from users
        for (i = 0; i < num; i++)
        {
            proc[i].index = i
            System.out.println("Enter the id of process : ");
            proc[i].id = in.nextInt();
            proc[i].state = "active";
            proc[i].f = 0;
        }
        // sorting the processes from on the basis of id
        for (i = 0; i < num - 1; i++)
        {
            for (j = 0; j < num - 1; j++)
            {
                if (proc[j].id > proc[j + 1].id)
                {

```

```

temp = proc[j].id;
proc[j].id = proc[j + 1].id;
proc[j + 1].id = temp;
}
}
}
for (i = 0; i < num; i++)
{
System.out.print("[" + i + "]" + " " + proc[i].id);
}

int init;
int ch;
int temp1;
int temp2;
int ch1;
int arr[] = new int[10];
proc[num - 1].state = "inactive";
System.out.println("\n process " + proc[num - 1].id + "select as co-ordinator");
while (true)
{
System.out.println("\n 1.election 2.quit ");
ch = in.nextInt();
for (i = 0; i < num; i++)
{
proc[i].f = 0;
}
switch (ch)
{
case 1:
System.out.println("\n Enter the Process number who initialsied
election : ");

```

```

init = in.nextInt();
temp2 = init;
temp1 = init + 1;
i = 0;
while (temp2 != temp1)
{
if ("active".equals(proc[temp1].state) && proc[temp1].f == 0)
{
System.out.println("\nProcess " + proc[init].id + "
send message to " + proc[temp1].id);
proc[temp1].f = 1;
init = temp1;
arr[i] = proc[temp1].id;
i++;
}
if (temp1 == num)
{
temp1 = 0;
}
else
{
temp1++;
}
}
System.out.println("\nProcess " + proc[init].id + " send message to " +
proc[temp1].id);
arr[i] = proc[temp1].id;
i++;
int max = -1;
// finding maximum for co-ordinator selection
for (j = 0; j < i; j++)

```

```

{
    if (max < arr[j])
    {
        max = arr[j];
    }
}

// co-ordinator is found then printing on console
System.out.println("\n process " + max + "select as co-ordinator");

for (i = 0; i < num; i++)
{
    if (proc[i].id == max)
    {
        proc[i].state = "inactive";
    }
}

break;

case 2:

    System.out.println("Program terminated ...");

    return ;

default:

    System.out.println("\n invalid response \n");

    break;

}

}

}

}

class Rr
{
    public int index; // to store the index of process

    public int id; // to store id/name of process

    public int f;String state; // indiactes whether active or inactive state of node

```

}

```
Activities Terminal Apr 6 5:43 PM
Terminal
3.2.3.jar: command not found
it@lt-Vostro-3710:~$ cd Documents
it@lt-Vostro-3710:~/Documents$ cd Ishwar1
it@lt-Vostro-3710:~/Documents/Ishwar1$ javac *.java
it@lt-Vostro-3710:~/Documents/Ishwar1$ java Ring
Enter the number of process :
5
Enter the id of process :
1
Enter the id of process :
2
Enter the id of process :
3
Enter the id of process :
4
Enter the id of process :
5
[0] 1 [1] 2 [2] 3 [3] 4 [4] 5
Process 5select as co-ordinator
1.election 2.quit
1
Enter the Process number who initialsted election :
2
Process 3send message to 4
Process 4send message to 1
Process 1send message to 2
Process 2 send message to 3
process 4select as co-ordinator
1.election 2.quit
2
Program terminated ...
it@lt-Vostro-3710:~/Documents/Ishwar1$

Terminal
3.2.3.jar: command not found
it@lt-Vostro-3710:~$ cd Documents
it@lt-Vostro-3710:~/Documents$ cd Ishwar1
it@lt-Vostro-3710:~/Documents/Ishwar1$ javac *.java
it@lt-Vostro-3710:~/Documents/Ishwar1$ java Bully.java
Error: Could not find or load main class Bully.java
it@lt-Vostro-3710:~/Documents/Ishwar1$ java Bully
3 Active process are:
Process up = p1 p2 p3 p4 p5
Process 5 is coordinator
.....
1 up a process.
2.down a process
3 send a message
4.Exit
2
Bring proces up
3
process3is already up
.....
1 up a process.
2.down a process
3 send a message
4.Exit
2
Bring down any process.
1
.....
1 up a process.
2.down a process
3 send a message
4.Exit
3
which process will send message
3
ok
.....
1 up a process.
2.down a process
3 send a message
4.Exit
4
it@lt-Vostro-3710:~/Documents/Ishwar1$
```